

Addison Alvey-Blanco

aja13@illinois.edu · github.com/a-alveyblanc

Technical Summary

Compiler and high-performance computing engineer specializing in high-level program transformations for numerical applications. Experience with MLIR-based compiler passes, polyhedral transformations, dependence analysis, GPU-oriented code generation, kernel fusion, tensor contraction optimization, and transformation pipelines for high-order numerical methods.

Education

University of Illinois Urbana-Champaign

Aug 2022 – Present

Ph.D. in Computer Science, expected 2027

Advisor: Andreas Klöckner

Working dissertation title: High-Level Code Transformations for Numerical Applications

Northern Illinois University

Aug 2019 – May 2022

B.S. in Computer Science & Mathematics

Experience

Sandia National Laboratories

May 2024 – Present

Year-Round Intern

- Developing a portable abstraction for hardware-accelerated matrix multiplication intrinsics
- Designed and implemented a platform-agnostic kernel fusion pipeline in an MLIR-based compiler, reducing intermediate memory traffic across fused linear algebra computations
- Designed and implemented an algebraic contraction-reordering compiler pass for fused kernels, reducing operation counts and temporary storage requirements when profitable
- Contributed compiler infrastructure for representing, transforming, and lowering high-level linear algebra computations toward accelerator-oriented implementations

National Center for Supercomputing Applications (NCSA)

Aug 2022 – Oct 2026

Research Assistant, Center for Exascale-Enabled Scramjet Design (CEESD)

- Developed compiler transformations for numerical applications targeting high-performance CPU and GPU architectures
- Developed compiler infrastructure for tensor contractions, array expressions, temporary storage, dependence analysis, and code-generation transformations
- Implemented computational-graph rewrites exploiting tensor-product structure to reduce run-time cost and intermediate storage in discontinuous Galerkin operator application

Selected Compiler Projects

Compiler-Controlled Array Materialization and Dependence Analysis

- Developing a polyhedral compiler transformation for explicitly materializing array expressions into temporary storage, providing control over where and when intermediate values are computed
- Designed placement relations that map logical expression instances to physical buffer locations

and execution contexts, supporting tiling, storage reuse, and rolling-buffer realizations

- Developed dependence-analysis machinery that reconstructs precise producer–consumer relationships after materialization and storage remapping
- Developing scheduling and verification machinery for transformed programs whose coarse dependence graphs may contain cycles while precise instance-level dependencies admit valid execution orders

Tensor-Product Numerical Transformation Pipeline

- Building a domain-specific transformation pipeline for tensor-product discretizations that rewrites high-level numerical expressions into structured computations suitable for GPU code generation
- Implemented computational-graph transformations that expose tensor-product structure and reduce arithmetic work and intermediate storage
- Developing algebraic and storage transformations for tensor contractions, operator application, and intermediate materialization before low-level scheduling and code generation
- Applying the pipeline to discontinuous Galerkin solvers for high-order numerical simulation

MLIR Kernel Fusion and Algebraic Optimization

- Developed a platform-agnostic MLIR kernel fusion pipeline that combines producer–consumer linear algebra operations and avoids unnecessary intermediate materialization
- Developed algebraic reordering transformations for fused tensor contractions to reduce asymptotic operation counts and temporary storage requirements
- Investigated portable lowering of fused kernels to hardware-accelerated matrix multiplication operations

Publications

- **A. Alvey-Blanco**, B. Kelley, S. Rajamanickam, “Platform-Agnostic Kernel Fusion Pipeline for Linear Algebra Kernels in MLIR,” LLVM-HPC 2026, SC26, *Accepted, to appear*.
- A. Klöckner, A. Fikl, X. Wei, T. Gibson, **A. Alvey-Blanco**, “modepy: Basis Functions, Interpolation, and Quadrature (not just) for Finite Elements,” *Journal of Open Source Software*, 2026. DOI: 10.21105/joss.09294
- **A. Alvey-Blanco**, B. Kelley, S. Rajamanickam, “Platform-Agnostic Kernel Fusion Pipeline for Linear Algebra Kernels in MLIR,” in *Computer Science Research Institute Summer Proceedings 2025*, Sandia National Laboratories, 2025. *Preliminary version; extended version accepted to LLVM-HPC 2026*.
- **A. Alvey-Blanco**, K. Liegeois, B. Kelley, “Toward Automatic Kernel Fusion for Kokkos Using MLIR,” in *Computer Science Research Institute Summer Proceedings 2024*, Sandia National Laboratories Technical Report SAND2024-16688O, 2024, pp. 204–215.

Technical Skills

Compilers: MLIR, LLVM, polyhedral compilation, isl/islpy, dependence analysis, computational-graph transformations

HPC: GPU code generation, kernel fusion, tensor contractions, accelerator programming, performance optimization

Numerical Computing: Discontinuous Galerkin methods, finite elements, tensor-product dis-

cretizations, numerical linear algebra

Programming: Python, C++

Teaching Experience

Numerical Analysis (CS 450, UIUC)

Fall 2026

Incoming Graduate Teaching Assistant

Numerical Analysis (CS 450, UIUC)

Fall 2023

Graduate Teaching Assistant

Databases (CS 466, NIU)

Spring 2022

Undergraduate Teaching Assistant

Leadership

Treasurer, SIAM UIUC Student Chapter

Spring 2024 – Spring 2026